

CLAUDE.md

This file provides guidance to Claude Code (claude.ai/code) when working with code in this repository.

What this is

A personal indoor-cycling training log (Christopeit AX 4000 + Kinomap), not a software project. No build system, no package manager — just a single ground-truth CSV, a static HTML dashboard that reads it, and a PDF export of that dashboard.

Architecture: one CSV, everything else derived

training_log.csv is the **only** hand-maintained data in this repo. training.html is static — it fetch()es the CSV at load time, parses it client-side, and derives every chart, table row, hero metric, weekly rollup and legend from it at render time. Adding a session means appending one row to the CSV (via the add-training skill) — training.html itself never needs to be edited.

- training_log.csv — one row per session, or per rest-day BP-only log. Columns:
 - Datum (D.M, no year), Nr (session label — plain sequential integer since ~week 9, – for rest days; not derivable, always script-assigned)
 - Dauer_sek, Kcal, Watt (estimated, see formula below), MaxHF, Rec (editorial “strong session” flag, 1 or blank)
 - RR_ruhe, RR_training, RR_abend — blood pressure readings as SYS/DIA/PULS, optionally with a trailing (HH:MM)/(Mittag) annotation. Rest-day rows carry real BP readings too — Nr: ‘-’ only means “no training that day”, not “no data”.
 - Kinomap, Strecke, Distanz_km, Kadenz_rpm, Watt_Max, Watt_Max_PR (☐ — set automatically when a session’s max watt beats every prior Watt_Max in the file), Hoehenmeter, Temperatur_C, Max_kmh — structured fields parsed out of the Kinomap on-screen summary; blank for non-Kinomap sessions.
 - Kommentar — free-text remainder (Stufentest/Pyramide labels, warnings, subjective notes) that doesn’t fit a structured column above.
 - **Not stored — always computed by training.html at render time:** weekday, week number ($\text{Math.floor}((\text{date}-25.5)/7)+1$, fixed 7-day blocks from 25.5, not ISO weeks), cumulative kcal, week colors (cycled from a fixed palette), week date-range labels, hero metrics, and every chart’s axis bounds. There is nothing to keep in sync by hand — if a

number ever looks stale, it’s a bug in training.html’s derivation, not a missed manual edit.

- training.html — reads training_log.csv via fetch() and renders all 4 tabs (Täglich/Wöchentlich/Tabelle/Plan 200W) with Chart.js (loaded from cdnjs.cloudflare.com, no local deps). Because it does a real fetch(), it must be served over HTTP (python3 -m http.server from the repo root, then open http://localhost:<port>/training.html) — opening the file directly (file://) fails silently because Chrome blocks local-file fetch() under file:// (CORS). The Plan tab’s phase-progress prose cards and the “Ziel” (target watt) series are a hand-authored training plan, not logged data — those stay manually edited and are unaffected by any of this.
- training.pdf — a print export of training.html (all four tabs, light theme, landscape, pure white background), regenerated by hand after editing the CSV — see recipe below. Not needed after a normal add-training run, which regenerates it automatically.
- stufentest.html / stufentest_regression.html — standalone step-test (Stufentest) watt/HR calibration charts, unrelated data (watts[]/bpms[] arrays), used to derive the HR-based watt estimation formula. Untouched by the CSV refactor.
- wkg_histogram.html, training_slideshow.html — other standalone single-purpose dashboards with their own embedded data. Untouched.
- PulsCharts/ — a self-contained folder: puls_graphen_uebersicht.html plus the watch-screenshot .jpg files it displays, referenced by plain relative filename (no subfolder) in the .

Generating training.pdf

training.pdf must be regenerated by hand after editing training_log.csv directly (the add-training skill does this automatically — this recipe is for a manual/bulk edit). Because training.html fetches the CSV, it must be rendered through a local HTTP server, not file://.

Recipe (headless Google Chrome + poppler’s pdftoppm/pdftotext for verification):

```
SCRATCH=/path/to/scratch # any writable temp dir
python3 - <<'EOF'
p = '/Users/haraldbeker/training/training.html'
s = open(p).read()
s = s.replace(".sec{display:none}", ".sec{display:block}") # show all 4 t
s = s.replace("const isDark=matchMedia('(prefers-color-scheme: dark)').matc
open('/Users/haraldbeker/training/_training_print_tmp.html', 'w').write(s)
EOF

PORT=8756 # any free local port, not 5000 (macOS AirPlay)
```

```
(cd /Users/haraldbeker/training && python3 -m http.server $PORT >/dev/null & & sleep 1
```

```
"/Applications/Google Chrome.app/Contents/MacOS/Google Chrome" \
--headless --disable-gpu --no-pdf-header-footer --virtual-time-budget=8000 \
--print-to-pdf="$SCRATCH/final.pdf" "http://localhost:$PORT/_training_print_tmp.html"
```

```
cp "$SCRATCH/final.pdf" /Users/haraldbeker/training/training.pdf
rm /Users/haraldbeker/training/_training_print_tmp.html
lsof -ti:$PORT | xargs -r kill
```

Before trusting the output, verify at high DPI — the low-res inline PDF preview is not reliable enough to catch either of the bugs below:

```
pdftoppm -png -r 200 -f 1 -l 1 /Users/haraldbeker/training/training.pdf "$SCRATCH/check"
# crop/zoom the daily "Ø Watt pro Tag" chart and confirm the tallest bar's pixel
# height actually lines up with its real value's gridline.
```

Pitfalls already fixed in training.html — don't reintroduce them:

- **Chart bars/lines can render squashed relative to their own y-axis even when Chart.js's config and data are correct** (e.g. a value of 213 only reaching the "140" gridline on a 100–230 axis). Root cause: Chart.js's default ~1000ms bar-growth animation gets captured mid-animation by the headless PDF snapshot, since `--virtual-time-budget` doesn't reliably advance `requestAnimationFrame`. Fixed by `animation:false` in `baseOpts` (in the script's `render()` function) — every chart spreads `...baseOpts`, so this covers all 9 charts. If squashing ever comes back, check this line first.
- **Chart axis bounds are auto-derived from the actual data on every render** (`scaleMin/scaleMax` helpers, ~8% headroom, rounded to a clean step) — this is no longer a hand-maintained value that can silently clip real data past an old max, unlike before this refactor.
- **Print-only CSS overrides** live in `@media print{...}` near the top of `<style>` (pure white `--bg/--surface/--surface2`, `@page{size:landscape}`, `body{max-width:none}` so the wide session table fits) and later (`.rec-row td{background:none}` — suppresses the "sehr gute Leistung" row highlight in print to save toner; must be declared *after* the base `.rec-row td` rule, since two rules with equal CSS specificity resolve by source order even across separate `@media print` blocks).
- `@media print{.card{break-inside:avoid;page-break-inside:avoid}}` (next to `.card{...}`) stops a chart card from being sliced across a page boundary.
- `_training_print_tmp.html` must be written **inside the repo root** (next to `training_log.csv`), not in `$SCRATCH` — its relative `fetch('training_log.csv')` resolves against whatever URL it's served from.

Key domain facts

- Watt is estimated from kcal/min, not measured directly: $\text{Watt} = (\text{kcal/min} \times 1000 \times 4.185) / 60 \times 0.25$, i.e. $\text{Watt} = \text{kcal/min} \times 17.4375$. Inverting it: $\text{Dauer_min} = \text{kcal} \times 17.4375 / \text{Watt}$. Displayed with a `~` prefix everywhere, since it's always an estimate.
- Dates are D.M with no year (implicit 2026), no leading zeros (e.g. '5.6', '14.8').
- Training weeks are fixed calendar boundaries starting Monday-ish from 25.5, 7-day blocks, not ISO weeks — computed by `getWeek()` in `training.html`'s script, not stored anywhere.
- A day can have multiple sessions (e.g. two rides logged the same date); when aggregating per day, sum kcal and average watt across that day's entries (see `dayMap` in `training.html`'s `render()` function).
- The "Total Dauer" hero/footer figure truncates each session's seconds before summing (not rounding the grand total) — a quirk preserved from the pre-CSV data so the displayed total didn't shift during the migration.